FastSpeech 2 Text-to-Speech for English and Telugu

Technical report. Repository: github.com/23f2001033/IITH-Intern-Assignment-

1. Summary

I trained two separate FastSpeech 2 models from the ming024/FastSpeech2 implementation: a multi-speaker English model on Svarah and a single-speaker Telugu model on IndicTTS Telugu. Each used about 5 hours of training audio and 1 hour of validation audio, selected by a fixed seed, with no clip, recording or transcript shared between the splits. Both models use the universal HiFi-GAN vocoder at 22,050 Hz.

Both models produce intelligible speech for all ten test sentences, and the text normalisation (times, dates, ordinals, AM) works in both languages. On 60 held-out English sentences, Whisper's word error rate is 22% for the final English model against 14% for the same sentences resynthesised from their true mel-spectrograms, while predicted naturalness (UTMOS) is about the same (2.63 against 2.65). Most of the remaining loss is therefore intelligibility lost in the acoustic model, not in the vocoder. Three native Telugu speakers rated the Telugu output as clear (clarity 5.0 of 5 from the reviewer who scored every clip) but somewhat flat (naturalness 3.6). The main failures are in prosody: English lexical stress is not realised even when the phonemes are right (the "conduct" homograph), split words are not recovered in English, and final vowels were too short and quiet because the training clips were trimmed to the first and last phone. Section 7.4 describes the inference-time correction for the last problem, and Section 9 the proper fix.

2. Approach and changes to the reference code

The model architecture is unchanged. The work went into the parts around the model that had to change for these datasets and for the test sentences. `core/CHANGES.md` lists every change; the important ones are below.

2.1 Text frontends

The test sentences probe four things in each language: a long or rare word, a homograph, a malformed word and text normalisation. The frontend decides most of these before the model sees anything, so I treated it as part of the system.

- **English.** The reference looks each word up in a fixed lexicon, which gives "conduct" one pronunciation in both E2a and E2b. I run grapheme-to-phoneme conversion on the whole sentence with part-of-speech tags, so the noun becomes `K AA1 N D AH0 K T` and the verb `K AA0 N D AH1 K T`. I also expand times, dates and symbols, read AM and PM as letters, rejoin clitics such as `'s`, and added a pronunciation for the code-mixed word "andaroo". The reference frontend is kept as `--frontend stock` so that the two can be compared on the same model.
- **Telugu.** ming024 has no Telugu support. Telugu spelling is close to phonemic, so I wrote a rule-based G2P covering the inherent vowel, vowel signs, virama, geminates, place-assimilated anusvara and visarga, with 46 phones. It also expands numbers, ordinals and times into Telugu words.

2.2 Alignment

FastSpeech 2 needs a duration for every phoneme, which comes from Montreal Forced Aligner (MFA) 3.4.3. English uses the pretrained `english_us_arpa` acoustic model. No pretrained Telugu model exists, so I trained one on the selected Telugu audio with my G2P lexicon and then aligned the same audio with it.

2.3 Bugs fixed in the reference code

Problem	Effect if left
Blank TextGrid intervals were dropped, and current MFA writes silence as blanks	Pauses lose their frames, so durations no longer match the mel-spectrogram
Unknown phones were silently removed	Phone and duration sequences go out of step
The energy predictor received `p_control` instead of `e_control`	Energy control had no effect, and pitch control scaled energy
Energy was the only target not cast to float32	With NumPy 2, training crashes on a GPU with a dtype error, while CPU runs pass
Hard-coded `.cuda()` calls in the STFT code	Feature extraction crashes without a GPU

2.4 Other engineering changes

- The split uses fixed id lists instead of a random validation size, and normalisation statistics come from the training set only.
- The length regulator is vectorised, because the original forced a GPU synchronisation per phoneme. Output was identical on 200 random cases.
- Training tracks the best validation checkpoint and stops cleanly before Kaggle's 12-hour limit.
- Training and inference stop if the mel settings differ from HiFi-GAN's configuration.

3. Data selection, split and configuration

3.1 Selection

1. Keep clips of 1 to 15 seconds whose transcripts can be read from the text. English clips containing long digit strings such as application numbers are rejected, because their spoken form cannot be recovered reliably.
2. Resample to 22,050 Hz, the rate HiFi-GAN expects, and normalise peak level.
3. Drop speaking-rate outliers (outside the 1st to 99th percentile), which usually indicate a bad transcript. For English, also drop the noisiest 15% by estimated SNR, because Svarah is not studio audio.
4. Group clips sharing a recording or transcript, so that a group always lands in one split.
5. Shuffle groups with seed 1234 and fill validation, then training, by cumulative duration.

The targets are 5.2 h and 1.05 h, slightly above 5 h and 1 h, because the aligner trims leading and trailing silence.

Speakers. Svarah has no speaker column, so I identified speakers from the combination of metadata fields, which gives 115 groups for its 117 speakers. This is a proxy: two speakers with identical metadata would be merged. English speakers deliberately appear in both splits, because a speaker embedding only learns from training clips. Telugu has two voices recorded at different rates. I used gender id 1 only,

which has 12.79 h at 48 kHz, enough for the full selection without mixing voices or upsampling 16 kHz audio. The dataset card's figure of 8.74 h did not match what I measured. The dataset labels this voice male, but its median F0 of 186 Hz is high for an adult male. All three native Telugu reviewers (Section 7.2) independently described the generated voice as male, so I keep the dataset's label.

3.2 Result

	English (Svarah)	**Telugu (IndicTTS)**
Clips in the dataset	6,656 (9.61 h)	8,576 (36.71 h); selected voice 4,533 (12.79 h)
Source sample rate	16 kHz	48 kHz (selected voice)
Rejected: other voice	Not applicable	4,043
Rejected: duration	1,476	727
Rejected: text	223	11
Rejected: speaking rate	100	76
Rejected: SNR	729	Not applied
Train: clips, hours	3,304, 5.216 h	2,417, 5.201 h
Validation: clips, hours	692, 1.061 h	505, 1.051 h
Train after silence trim	3,303 clips, 4.846 h	2,417 clips, 4.428 h
Validation after silence trim	692 clips, 0.981 h	505 clips, 0.890 h
Speakers (train, validation)	112, 98	1, 1
Median clip length (train)	5.07 s	7.74 s
Median estimated SNR (train)	32.2 dB	51.2 dB
Median F0 (40-clip sample)	143.8 Hz	186.2 Hz
Lexicon words	6,805, of which 1,195 came from neural G2P	13,447, all from the rule-based G2P

The English training set has 3.01 h of female and 2.21 h of male speech; validation has 0.46 h and 0.60 h. All 3,996 selected English clips aligned, and one was skipped as unusable during feature extraction. All 2,922 selected Telugu clips aligned, and none were skipped.

Effect of silence trimming. The selected audio meets the targets, but the hours the model trains on are lower because leading and trailing silence is removed. English loses 7%, leaving 4.85 h. Telugu loses 15%, leaving 4.43 h for training and 0.89 h for validation, because its clips carry more silence at the edges than the margin I allowed for. Selecting more audio would have meant repeating about four hours of alignment, so I kept this split and report both figures.

Leakage checks. `verify_split()` stops the run unless clip, transcript and recording overlap between the splits are all zero, no test sentence appears in the data, every validation speaker is in training, and every file is at 22,050 Hz. All checks passed for both languages.

3.3 Model and training configuration

Setting	Value
Encoder and decoder	4 and 6 layers, hidden size 256, 2 heads, convolution filter size 1024
Variance predictors	Filter size 256, kernel 3, dropout 0.5; 256 linear pitch and energy bins
Pitch and energy	Phoneme level, normalised with training-set statistics
Audio features	22,050 Hz, FFT 1024, hop 256, window 1024, 80 mels, 0 to 8,000 Hz
Speakers	English: speaker embeddings. Telugu: none
Optimiser	Adam, betas 0.9 and 0.98, batch size 16, gradient clip 1.0, 4,000 warm-up steps
Hardware	One NVIDIA T4 per model on Kaggle, 10-hour limit per run
English schedule	About 90 steps a minute. Stopped at the 10-hour limit at step 54,153, so it reached only the first planned learning-rate drop (x0.3) at step 50,000. Lowest validation total loss at step 11,000
Telugu schedule	32,000 steps in 5 h 44 min (about 93 steps a minute), with learning-rate drops (x0.3) at 16,000, 22,400 and 28,800. Lowest validation total loss at step 13,000

The schedules differ because I measured English speed only after that run started, and restarting it would have cost more than the later drops were worth. I sized Telugu from an early measurement, expecting its longer clips to run more slowly. It ran at a similar speed, so it finished in 5 h 44 min, under the limit. Both still use the inverse-square-root decay after warm-up, but this difference has to be kept in mind in the comparison in Section 8.

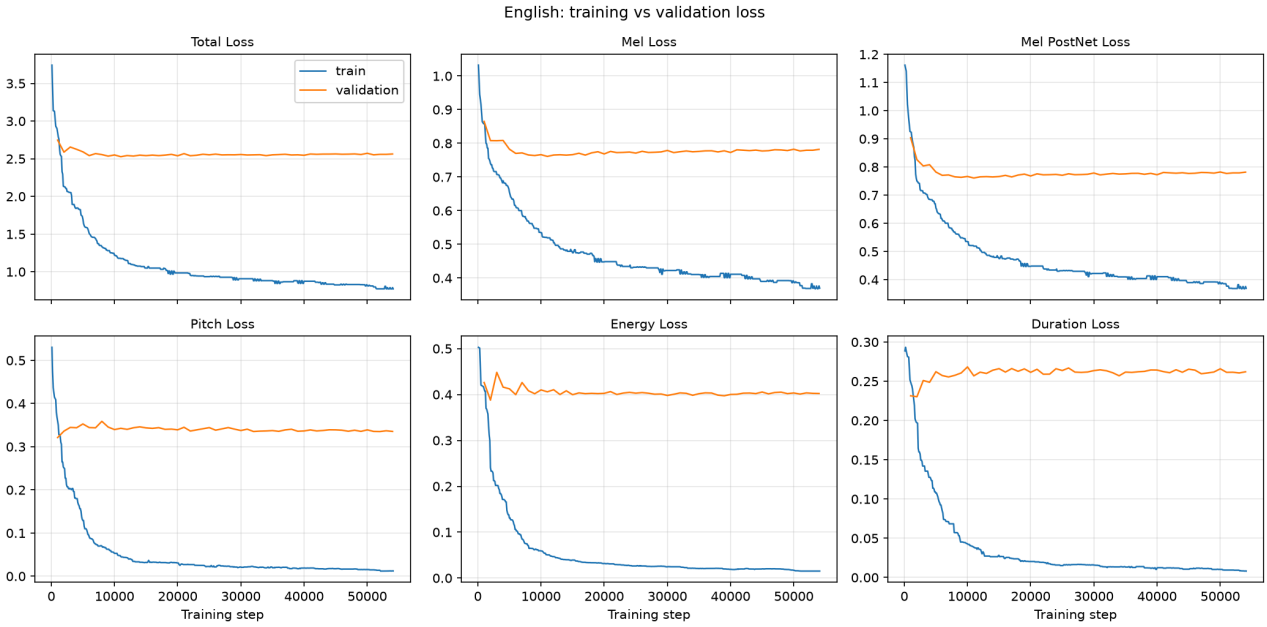
Finishing the English schedule. After evaluation (Section 7) showed that the late English checkpoint was better than the best-validation one, I resumed English from step 54,153 with its optimizer state and ran it to step 60,000 (`notebooks/04_finish_english.ipynb`). Because the remaining budget was short, the later learning-rate drops were moved to steps 55,000 and 57,500 with the new `--anneal_steps` option, so the run ends with all three drops like Telugu. The validation total at step 60,000 was 2.561.

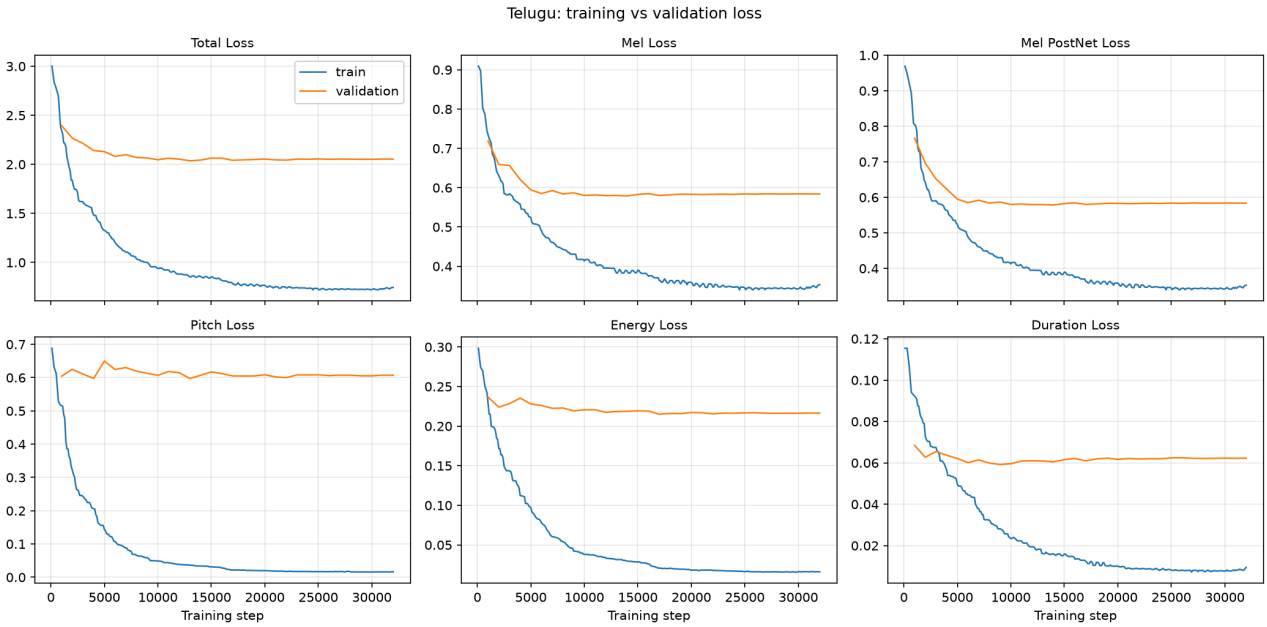
Checkpoints used for the test sentences. English: step 60,000, voice `spk_f0bb6d4e`. Telugu: step 13,000, the lowest validation loss. Section 7.3 gives the reasons and how strong the evidence is.

4. Loss curves

Validation here is teacher-forced: the model receives the true durations, pitch and energy. Validation loss therefore measures how well the model fits held-out clips given correct prosody, and it does not directly measure free synthesis.

Training curves are in `Results/<Language>/training_loss.png` and validation curves in `validation_loss.png`. The figures below overlay the two. Training loss is logged from one batch every 100 steps, so its line is a rolling median of 20 logged points. Validation loss is computed on the full validation set every 1,000 steps.





Validation loss	Step	Total	Mel	Pitch	Energy	Duration
English, first check	1,000	2.748	0.865	0.321	0.426	0.232
English, lowest total	11,000	2.528	0.761	0.343	0.406	0.257
English, last check	54,000	2.564	0.782	0.335	0.402	0.262
Telugu, first check	1,000	2.395	0.719	0.604	0.237	0.069
Telugu, lowest total	13,000	2.037	0.580	0.597	0.218	0.061
Telugu, last check	32,000	2.053	0.584	0.607	0.216	0.062

Convergence. Both models trained stably, with no divergence or skipped batches recorded in the logs. Training loss kept falling to the end of each run, reaching a median total of 0.79 for English and 0.73 for Telugu over the last 20 logged points.

Train-validation gap. Validation stopped improving much earlier than training. The lowest validation total came at step 11,000 for English and step 13,000 for Telugu, both before any learning-rate drop. After those points, lower training loss brought no validation gain, so the later training mostly fitted the training clips. The gap is largest for the variance predictors: at the end of training, Telugu pitch loss is about 0.016 on training batches and 0.607 on validation, and English duration loss is about 0.009 and 0.262.

Overfitting. Both models overfit after their best step, in different ways. Telugu memorises its training data without validation getting worse: training pitch loss falls to 0.016 while validation pitch stays at about 0.60, a gap of almost 40 times, yet the validation total stays between 2.04 and 2.06 after step 10,000. English validation mel loss rises slowly from 0.761 to 0.782, and the learning-rate drop at step 50,000 lowers training loss without lowering validation loss. In both languages the change in validation total after the best step is under 2%, so on this measure the best step is only slightly better than the final one. Section 7 shows that this measure did not pick the better English checkpoint.

Variance predictors. Validation pitch and duration loss barely improved after the first check at step 1,000, while training values fell by more than an order of magnitude. A likely reason is that text alone only partly determines intonation and timing, so what generalises is learned early and the rest is memorised. Stronger regularisation (higher predictor dropout, or early stopping of the predictors alone) is the obvious next experiment, which I did not run. English also has many accents in one model and alignments from a US English acoustic model, which may add noise to its duration targets. Absolute losses are not comparable between the languages, because the data, speakers and normalisation differ.

5. Speech observations per test sentence

Test text is kept exactly as given. Normalisation happens inside the frontend only. The audio is in `Results/<Language>/test_generated_audios`. Sources for each note: English listening notes are mine (I do not speak Telugu), made at step 54,153. Telugu notes come from the three native reviewers in Section 7.2, who heard step 13,000 before the playback correction in Section 7.4. Durations and pitch come from the final models' predicted values in `variance_adaptor_plots/values`.

Id	Sentence	Probe	Observation
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	Very long rare word	All 28 phones of the long word are produced, over 1.99 s at the same average rate as the rest of the sentence (71 ms a phone), and the word is intelligible. Its pitch is nearly flat (mostly 185–215 Hz) and the primary stress gets no pitch accent, so it sounds monotone. The first word "the" was hard to catch in earlier versions.
E2a	The orchestra's conduct during the ceremony was praised by the audience.	Homograph, noun	The frontend gives the noun pronunciation `K AA1 N D AH0 K T`, but the word sounded closer to con-DUCT. The second vowel is predicted as long as the first and louder (Section 6).
E2b	The researchers will conduct an experiment to evaluate the proposed method.	Homograph, verb	The frontend gives the verb pronunciation `K AA0 N D AH1 K T`, but the word sounded like CON-duct. The stressed second vowel is predicted shorter and lower than the first. The two senses get different phonemes but, in effect, reversed stress.

E3	The rhyth m of the orchestra changed unexpectedly.	Malformed word, kept split	Not recovered. G2P reads "rhyth" as `R AY1 DH` and the lone "m" as the letter name `EH1 M` ("em"), so the listener hears two words, the second with the most energy in the sentence. The reference frontend makes the same error.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	Time, date, code-mixing	Read as "ten forty five A M on the twelfth of November twenty twenty six". The date was clearly heard. The comma gives a 0.30 s pause. "andaroo" uses the added pronunciation `AA1 N D AH0 R UW1`. The last few words were less clear than the start.
T1	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.	Long rare word	Reviewer clarity 5 of 5, and the first word was "spoken full and correctly". The long word takes 1.46 s and carries the sentence's pitch peak. The last word was heard as cut: its final vowel was predicted at only 58 ms and near-zero energy.
T2a	తాడుపై అతని పట్టు చాలా బలంగా ఉంది.	Homograph, grip	Clear (5 of 5), but one reviewer found it "a bit rushed". It is the fastest of the full sentences at 15.1 phones a second. One reviewer reported the first word missing (see Section 7.4).
T2b	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.	Homograph, silk	Clear and fairly natural (4 of 5). The reviewer heard a difference in పట్టు between T2a and T2b, which can only come from prosody (Section 6). The last word was heard as cut.
T3	పులి అడ విలో నిశ్శబ్దంగా నడిచింది.	Malformed word, kept split	Heard as the intended word అడవిలో, with no break. Clarity was 5 of 5 at step 13,000. At step 32,000 the reviewer found నిశ్శబ్దంగా mispronounced.
T4	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.	Year, ordinal, time	The numbers are read as రెండు వేల ఇరవై ఆరు, పన్నెండవ and పది గంటల నలభై ఐదు నిమిషాలకు. The reviewer found the time "very stressed", the intonation flat (naturalness 3 of 5), and some background noise in the 0.56 s comma pause. The first word ఈ is only 151 ms long, and one reviewer reported it missing.

Effect of the English frontend. `Results/English/stock_frontend_comparison` holds the same sentences through the reference frontend (step 11,000, voice `spk_d270faea`). Its phonemes show the differences directly:

- **"conduct":** the reference gives `K AA1 N D AH0 K T` in both sentences.
- **"10:45 AM":** it becomes `T EH1 N T ER0 F AY1 T IH0 V AE1 M`, with the minutes garbled and AM read as the word "am".
- **"2026":** it ends in `S K AY1 Z` instead of "six".
- **"andaroo":** it becomes `AE0 N D AA1 R OW0`.

The improved frontend fixes all four, so the probes for time, date and code-mixing depend on the frontend far more than on the model. It does not fix the split word in E3.

6. Variance adaptor analysis per test sentence

For every sentence, `Results/<Language>/variance_adaptor_plots/{duration,pitch,energy}/<Id>.png` plot the predicted duration, pitch (in Hz) and energy for each phoneme, with word boundaries marked. `values/<Id>.json` holds the same numbers. The values below are from the final models. Pitch and energy are predicted per phoneme from the encoder output before the length regulator. The final-vowel correction in Section 7.4 changes only the last phoneme's duration.

E1. Duration is close to uniform: the long word averages 71 ms a phone, the same as the sentence. Sibilants (`S` 104–116 ms, `SH` 139 ms) are the longest phones, as in natural speech. Pitch stays between 178 and 242 Hz across the word, with its single peak on the unstressed third syllable rather than on the primary stress `EH1` (192 Hz). Energy follows the phone class, high on vowels (60–96) and low on stops (9–22). That shows the predictor has learned segment-level energy, but it marks no word-level prominence. The sentence ends without a clear fall (about 180 Hz on "proceedings"), which matches the flat delivery heard.

E2a and E2b, the homograph. The input differs (`AA1 … AH0` for the noun, `AA0 … AH1` for the verb), so the predictors could separate them.

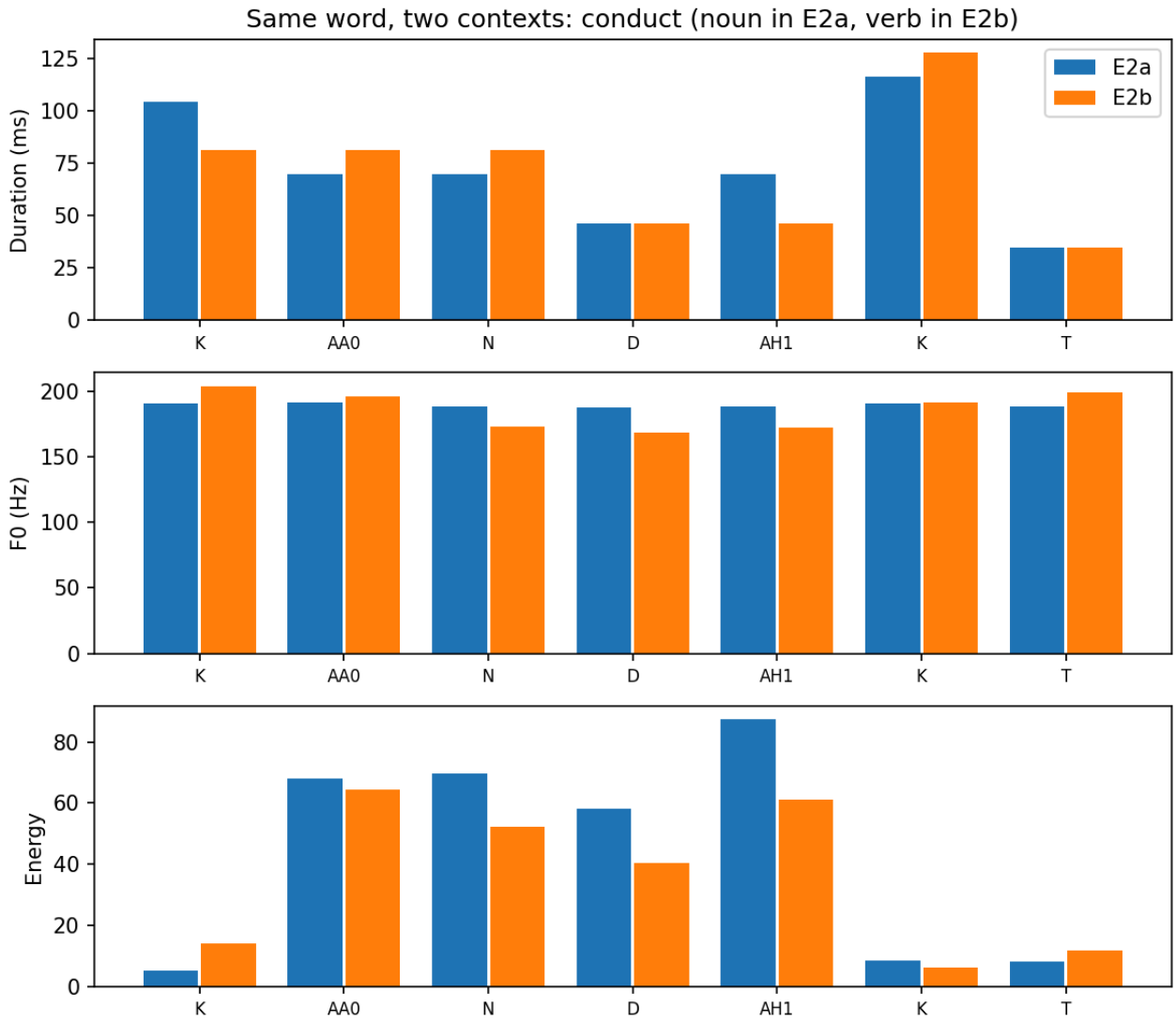
- **Noun:** the two vowels get equal durations (70 and 70 ms) and nearly equal pitch (192 and 189 Hz), and the unstressed second vowel is the louder (energy 88 against 68).
- **Verb:** the stressed second vowel is shorter (46 against 81 ms), lower (172 against 196 Hz) and slightly quieter (61 against 64) than the unstressed first.

Every cue goes against the lexicon, which explains why the verb was heard as CON-duct. The stress digit therefore reaches the model, but the predictors have not learned to turn it into duration, pitch or loudness.

Two causes are plausible, and I did not test either:

- The stress marks come from a US English lexicon, while Svarah speakers' Indian English often realises stress differently.
- The duration targets come from a US English aligner (Section 9.3).

`homograph_comparison.png` overlays the two words.



E3. "m" is treated as a separate stressed word: its vowel `EH1` gets the highest energy in the sentence (105) and a pitch rise to 217 Hz. The predictors faithfully render what the frontend produced, so the error comes from the frontend alone. The final `IY0` is 186 ms with low energy (27), a soft but complete ending.

E4. The sentence has the clearest English intonation:

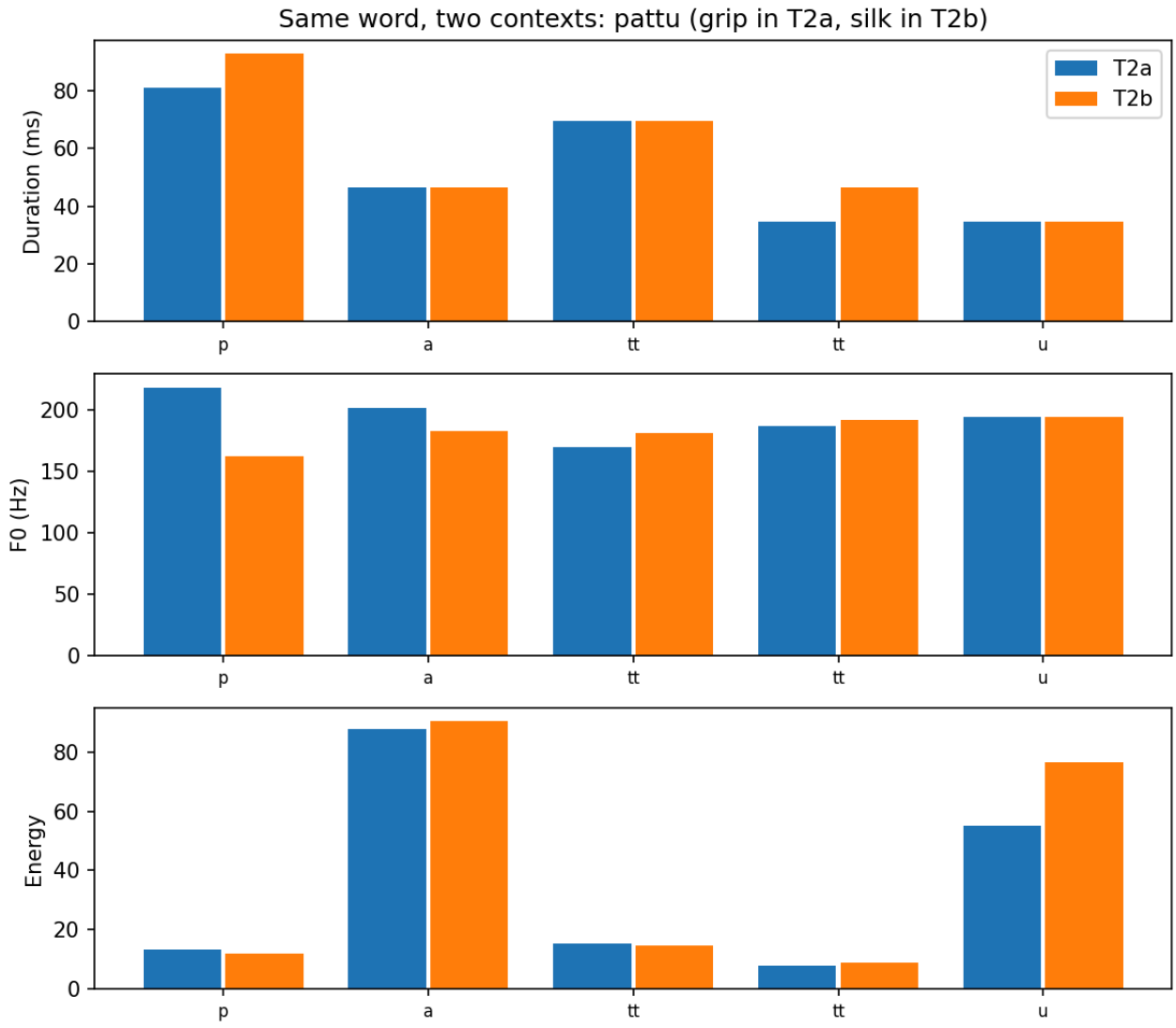
- Pitch starts high ("meeting" and "will" at about 246 Hz) and declines to 172 Hz on "time".
- The comma becomes a 302 ms `sp` token with near-zero energy (2.7), and pitch before and after it stays at about 203 Hz, a continuation contour.
- The expanded numbers get ordinary word durations ("ten" 209 ms, "forty" 348 ms, "five" 325 ms), and "andaroo" gets a long stressed first vowel (151 ms).

T1. The rare word gets the sentence's pitch peak (282 Hz on `oo`) and high vowel energy (up to 142), at a slightly slower rate than the rest (77 against 68 ms a phone). Pitch then declines steadily over the sentence, to 121 Hz at the end, a clear declarative contour. The final `u` is predicted at 58 ms with energy 4.8, against 60–140 on other vowels. The predictor learned this from training clips that end exactly at the aligner's last phone boundary, and it is why reviewers heard the last word as cut.

T2a and T2b, the homograph. పట్టు means "grip" and "silk" with the same spelling and the same sounds, so the phoneme input is identical: `p a tt tt u`. The predictions differ only through context.

పట్టు	Duration	Mean pitch	Mean energy	Geminate closure
T2a (grip, third word)	267 ms	194 Hz, falling from 219	36	105 ms
T2b (silk, second word)	290 ms	183 Hz, rising from 163	40	116 ms

The pitch shape differs: falling in T2a and rising in T2b, before the high-pitched చీరను. That is the difference the reviewer heard. The geminate is realised as a long closure in both, which keeps పట్టు distinct from a single `tt`. With no difference in the input, the model cannot express the meaning, and it should not need to. The overlay is in `Results/Telugu/variance_adaptor_plots/homograph_comparison.png`.



T3. Because the Telugu G2P works letter by letter, "అడ విలో" gives exactly the phonemes of అడవిలో (`a dd a v i l oo`), and no pause is inserted between words. The durations flow through the split: `a` 46 ms, then `v` 35 ms. This is why the reviewers heard one word. In English the same split fails, because G2P on a fragment is not the same as G2P on the whole word. T3 has the highest rate of the set (17.3 phones a second) and a final fall to 123 Hz.

T4. The sentence is 8.5 s long, with 128 phones.

- **Pause:** the comma gives a 557 ms pause with energy 0.3. Any noise heard there is produced by the vocoder from near-silent frames.

- **Pitch:** there are two declination phrases. Pitch falls to about 116 Hz at the end of the first clause, resets to 185–200 Hz after the comma, and falls again to 106 Hz.
- **The expanded time:** పది గంటల నలభై ఐదు has several vowels at energy 85–125 and an `ai` of 139–151 ms, longer than most vowels in the sentence. That is a plausible source of the "very stressed" impression.
- **First word:** ఈ is a single vowel of 151 ms.

7. Evaluation and choice of checkpoints

Listening alone could not separate the checkpoints or explain where quality is lost, so I added `scripts/evaluate_tts.py` (run by `notebooks/03_evaluate_speech.ipynb`). It uses 60 validation clips (seed 1234), never the test sentences, and produces each clip three ways:

- **Vocoder only:** the true mel-spectrogram through HiFi-GAN.
- **Teacher-forced:** the model given the true durations, pitch and energy.
- **Free:** the model predicting everything itself, which is how the test sentences are made.

Whisper large-v3-turbo transcribes each clip, giving a word error rate (WER). UTMOS (`utmos22_strong`) predicts naturalness on a 1 to 5 scale. Intervals are 95% bootstrap intervals over clips.

7.1 Objective scores

English (60 clips)	WER	UTMOS
Vocoder only (true mel)	0.145 [0.093, 0.205]	2.65 [2.53, 2.78]
Free, step 11,000 (best validation loss)	0.237	2.25
Free, step 54,153	0.227 [0.162, 0.294]	2.62 [2.50, 2.74]
Free, step 60,000 (final)	0.221 [0.162, 0.281]	2.63 [2.52, 2.75]
Teacher-forced, step 60,000	0.263 [0.196, 0.333]	2.36 [2.24, 2.49]

What the English numbers show:

1. **Naturalness is preserved, but intelligibility is lost in the acoustic model.** Free synthesis matches vocoder-only resynthesis on UTMOS (2.63 against 2.65), but its WER is 7.6 points higher. Vocoder-only audio already scores only 2.65. I did not score the original recordings, so I cannot say how much of that comes from the universal vocoder and how much from Svarah's recording conditions.
2. **True prosody does not help.** Teacher-forced synthesis is no more intelligible than free synthesis (the WER intervals overlap) and is clearly less natural. FastSpeech 2 trains its decoder with these true values, so this is not a train-test mismatch. Two plausible reasons: the true pitch and energy of conversational recordings are noisier than the predictors' averaged output, and the decoder of a 112-speaker model fits any one held-out clip only loosely. So for intelligibility, prediction error in the variance adaptor is not the main limit. This does not mean the predictors are good: Section 6 shows they miss lexical stress.
3. **Validation loss picked the wrong English checkpoint.** Step 11,000 had the lowest teacher-forced validation loss but is clearly less natural in free synthesis (2.25 against 2.63, intervals not overlapping). Steps 54,153 and 60,000 cannot be told apart. I ship 60,000 because it completes the learning-rate schedule, not because it scored better.

Telugu. Whisper cannot judge Telugu here: its WER is 0.82 even on vocoder-only resynthesis of real speech, and its character error rate is no better for real speech (0.43) than for synthesis (0.40). UTMOS was trained on English and separates nothing: 2.91 at step 13,000, 2.88 at 32,000 and 2.71 for vocoder only. For Telugu, only human listening counts.

7.2 Native Telugu listeners

Three native Telugu speakers received a blind A/B pack: each sentence at step 13,000 and at step 32,000, in randomised order, with a short rubric (clarity and naturalness from 1 to 5, a preference, and the probe questions). The key was not shared.

- **Reviewer 1** found all clips similar and clear.
- **Reviewer 2** heard no break in అడవిలో, could not tell A from B, heard the voice as male, and reported the first word missing in four of the five sentences.
- **Reviewer 3** scored every clip:

Reviewer 3	Clarity (mean of 5)	Naturalness (mean of 5)	Preferred
Step 13,000	5.0	3.6	T1, T2b, T3
Step 32,000	4.2	3.6	T2a

T4 was a tie ("both have defects"). Reviewer 3 also reported:

- the last word cut short in every clip;
- the time in T4 sounding over-stressed;
- flat intonation;
- background noise in the T4 pause, and a mispronounced నిశ్శబ్దంగా at step 32,000;
- the voice sounding male.

7.3 Checkpoint and voice choice

Telugu. The rule was the free-synthesis scores where they work and validation loss where they do not. For Telugu that gives step 13,000. Validation loss favours it by under 1%, and the one reviewer who could tell the versions apart preferred it in three of five sentences and gave it higher clarity. Two reviewers heard no difference. The evidence for 13,000 over 32,000 is therefore weak: the two are close, and 13,000 is the better-supported of two near-equal options.

English voice. Training used one speaker embedding per Svarah speaker, so the voice is chosen at synthesis time. I scored 12 voices on 30 validation clips each at step 60,000. The rule was the highest UTMOS among voices whose WER is below the median of the 12. That gives `spk_f0bb6d4e`, a female voice (WER 0.221, UTMOS 3.07 [2.89, 3.26]). The voice first used, `spk_d270faea`, has the most training audio but scored among the worst (WER 0.310, UTMOS 2.58), so the amount of data per speaker did not predict quality.

Caveats. With 30 clips, most WER differences between voices are within noise, and only the UTMOS differences are clear. I set the rule after seeing the scores. Before that, I had listened to test-sentence audio from a few voices. The final choice follows the validation scores, but that listening is a mild exposure to the test set and is disclosed here.

7.4 Correction applied to the generated audio

The reviewers' main complaints, the missing first word and the cut last word, have a measurable cause in the data preparation. Training clips were trimmed to the aligner's first and last phone. The model therefore learned to start speech on the first frame and to end on a short, fading final vowel.

- **The start.** In the generated audio, speech starts within 0–40 ms at full level. The first word is present in the waveform, but some players and messaging apps clip the start of playback. The first words reported missing are short: ఈ is 151 ms, పులి 197 ms, ఆమె 267 ms.
- **The end.** The final vowels were 58–93 ms long, at a small fraction of normal vowel energy.

Retraining with silence margins was not possible in the time left, so `core/synthesis.py` applies three corrections at inference time, to every test sentence in both languages:

1. **Final phone:** if the last phone is a vowel or sonorant shorter than 140 ms, its duration is raised to 140 ms and the decoder is run again with these durations. Stops and fricatives are left alone, so a final "d" is not stretched. In the final audio, the last phone was lengthened in all five Telugu sentences and in E4.
2. **Pauses:** the vocoder output inside predicted pause (`sp`) regions is lowered to 10%, with 15 ms ramps, to remove vocoder noise in silences.
3. **Padding:** the output gets 5 ms fades and 0.3 s of silence at each end.

These are corrections for a preprocessing mistake, not model improvements. Durations stated elsewhere in this report exclude the padding. The plots and `values` files show the durations actually used. I did not score the audio with and without the correction. Reviewer 3, who had reported the cut words, listened again to the corrected Telugu audio and confirmed that the first and last words are now complete. The proper fix is in Section 9.3.

8. English and Telugu compared

	English	Telugu
Task for the model	Many voices, varied Indian English accents, conversational recording conditions	One voice, recorded for speech synthesis
Pronunciation source	Stress-marked ARPAbet from a lexicon and neural G2P	Rule-based G2P, no stress marks
Alignment	US English acoustic model applied to Indian English	Model trained on the same audio it aligns, from about 6 h
Homograph	Resolved in the frontend by grammar	పట్టు has the same spelling and phonemes in both senses, so any difference is prosody only
Training speed	About 90 steps a minute	About 93 steps a minute

Losses. Telugu reaches lower validation losses on every term except pitch (total 2.04 against 2.56, mel 0.58 against 0.78, duration 0.06 against 0.26). Its duration loss is four times lower, which fits a single read-speech voice and an aligner trained on that same voice. English duration targets come from a mismatched aligner across 112 speakers. Telugu pitch is the hardest term in either language (validation 0.60), and it is also the most overfit. The schedules differed for most of training, because English had only its first learning-rate drop until the finishing run. But both reached their best validation loss before any drop, so the difference does not explain the gap.

Audio. Telugu sounds the more consistent of the two. The reviewers rated it clear, and its sentences carry clear declarative pitch falls. Its weak points are flat phrase-level intonation, an over-emphasised time expression, and the truncated endings, now corrected. English is less intelligible: a 22% WER on validation, against 14% for resynthesis. Its main failure is lexical stress, which the model receives but does not realise.

Probes.

- **Homograph:** English needs the frontend, because the two senses differ only in stress, and the model then fails to voice that difference. Telugu has no phonemic difference to express.
- **Split word:** Telugu's letter-by-letter G2P recovers it for free, while English G2P on fragments does not.
- **Normalisation:** it worked in both languages once handled in the frontend.
- **Long word:** both are rendered completely and intelligibly, but with flat word-level prosody.

9. Failures and improvements

9.1 Problems found while building the pipeline

- **Telugu alignment hang.** `mfa train` and `mfa align` share one database and MFA does not delete earlier alignments, so each clip held two overlapping alignments. The TextGrid export then failed an assertion, the worker died, and the main process waited indefinitely. I traced this in the MFA 3.4.3 source and fixed it by running with `--clean`. A watchdog now stops MFA if its working folder stops changing for 45 minutes. With the fix, the full Telugu run aligned all 2,922 clips.
- **GPU-only dtype crash.** The energy dtype bug in Section 2.3 did not appear in CPU testing, so it was found on the first GPU step. A short GPU test earlier would have caught it.

9.2 Failures in the generated speech

Failure	Evidence	Proposed improvement
Truncated starts and endings	Reviewers in every Telugu sentence; final vowels of 58–93 ms at near-zero energy; speech starting within 40 ms. Fixed in the submitted audio by the correction in 7.4, which the reviewer confirmed.	Keep 100–200 ms of silence at both ends when preparing the data, so the model learns real onsets and endings. The inference correction in 7.4 is a stopgap.
English lexical stress not realised	"conduct" heard with reversed stress; stressed vowels predicted shorter and lower (Section 6)	Better duration targets first (below). Then stress-aware supervision, such as a loss term or embedding that ties stress marks to duration and pitch, or Indian English stress labels.
English intelligibility below resynthesis	WER 0.221 against 0.145 for vocoder only	Better alignment, then more audio per speaker (below).
Split word not recovered in English	"rhyth m" read as "rithe em"	Before G2P, join adjacent fragments when the joined form is a dictionary word and the fragments are not. Never read a lone letter as a letter name inside such a join.

Flat intonation and an over-stressed time in Telugu	Reviewer naturalness 3.6 of 5; validation pitch loss stuck at 0.60 with a 40-fold train-validation gap	Regularise the pitch predictor (more dropout, earlier stopping), and use phrase-level pitch features, or a variational or flow-based prosody model, instead of pure regression to the mean.
Vocoder noise in pauses	Reviewer reported noise in the T4 pause	Fine-tune HiFi-GAN on the generated mel-spectrograms of each voice. For now, pauses are attenuated (7.4).

9.3 What I would change first

In order of expected effect on English:

1. **Adapt the aligner to Indian English.** The model's duration targets come from a US English acoustic model. English duration loss is four times that of Telugu, and the stress failure appears in the duration predictions first. Running MFA adaptation on the Svarah training audio before alignment is cheap and changes every target the model learns from.
2. **Choose the English data by speaker.** Spreading 5 h over 112 speakers gives under 0.2 h per voice. No Svarah speaker has more than about 0.19 h after filtering, so a few-speaker model would need far less than 5 h or a different corpus. Keeping fewer, cleaner speakers, ranked by recording quality, would trade accent coverage for a better-learned voice. The voice sweep (Section 7.3) shows that recording conditions matter more than the amount of audio per speaker.
3. **Keep silence margins when preparing the data**, as above.
4. **Fine-tune the vocoder.** I put this last, because free synthesis already matches vocoder-only resynthesis on naturalness, so the vocoder is not the main limit on these measures.

I would also score the original recordings alongside resynthesis, to separate the vocoder's losses from Svarah's recording quality.

9.4 Limitations

- Speaker identity in Svarah is inferred from metadata.
- The Telugu aligner, trained on about 6 h of audio, will be less precise than one trained on a large corpus, and I made no manual alignment check.
- Objective scores are English only. Whisper and UTMOS do not work on this Telugu audio, so the Telugu evidence rests on three listeners, one of whom scored every clip.
- The universal HiFi-GAN was not fine-tuned on these voices or on Telugu.
- The playback corrections in 7.4 were not scored with and without. They were checked by one reviewer re-listening.

10. Reproducibility

- The split uses seed 1234, and the exact clip ids are saved in the training and validation lists.
- The Kaggle notebooks used, in order:
  - `notebooks/01_prepare_data.ipynb`: data selection, alignment and features.
  - `02_train_and_generate.ipynb`: training and the test sentences.
  - `03_evaluate_speech.ipynb`: Section 7 and the final test-sentence audio.
  - `04_finish_english.ipynb`: English steps 54,153 to 60,000.

`README.md` gives the same steps as terminal commands.

- `requirements.txt` pins the Python packages, and MFA 3.4.3 is installed separately.
- Inference needs only a checkpoint and the small files in `English/assets` or `Telugu/assets`. The checkpoints (English step 60,000 and Telugu step 13,000) are at `https://huggingface.co/perceptron01/IIITH-FastSpeech2-English-Telugu`.
- The evaluation uses seed 1234 for clip selection and for the bootstrap.